

The Grid-Traffic Threshold: A Proof for Every $n \geq 496$

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Abstract

For integers $n \geq 2$ and $1 \leq a < n/2$, let

$$D(n) = \frac{1}{n} \binom{2n-2}{n-1}, \quad G(n, a) = \frac{n-2a+1}{n-a} \binom{n}{a} \binom{n-2}{a-1},$$

and put $R_n(a) = G(n, a)/D(n)$ and $\rho(n) = \max_{1 \leq a < n/2} R_n(a)$. Gil, Liang, Odetola, and Weiner conjectured that $\rho(n) < 1$ for every $n \geq 496$. We prove the conjecture. An adjacent-ratio polynomial localizes a row maximum exactly. Exact integer arithmetic then verifies all 99,505 rows with $496 \leq n \leq 100000$. For $n \geq 100001$, the same localization gives $(n - 2a_*)^2 < n$ at a maximizing index; explicit one-sided Stirling bounds reduce the remaining range to a rational endpoint estimate strictly below one. The finite and analytic parts use no floating-point inequalities.

1 Statement and context

The quantities above arise in the analysis of maximal traffic through lattice points when one point of a square grid is obstructed. Proposition 7.2 and Lemma 7.3 of Gil et al. [1] reduce their Conjecture 7.4 to the one-variable inequality stated in the abstract. They verify that inequality exactly for $496 \leq n \leq 2000$. The broad proof architecture used here—a uniform analytic tail joined to an exact finite certificate—has also been used in a different enumerative setting by Zeng [3]; no mathematical assertion in this article depends on that methodological comparison.

For a fixed n , the admissible set is

$$\mathcal{A}_n = \{1, \dots, \lfloor (n-1)/2 \rfloor\}.$$

It is finite and nonempty throughout the range considered, and every factor in the definition of $R_n(a)$ is positive. Our result is therefore equivalent to a strict inequality at every admissible pair.

Theorem 1. *For every integer $n \geq 496$,*

$$\rho(n) = \max_{a \in \mathcal{A}_n} R_n(a) < 1.$$

The proof is split at $n = 100000$. We first give a localization lemma used in both parts.

2 Exact localization of the row maximum

For $2 \leq a < n/2$, cancellation in the defining expression gives

$$q_n(a) := \frac{R_n(a-1)}{R_n(a)} = \frac{a(a-1)(n-2a+3)}{(n-a+1)^2(n-2a+1)}. \quad (1)$$

Set

$$H_n(t) = n^2 - n(2t^2 + 3t) - t - 1.$$

Lemma 2 (Row localization). *For every $n \geq 496$, let a_0 be the first integer $2 \leq a_0 < n/2$ for which*

$$H_n(n - 2a_0 + 1) \geq 0.$$

Then $a_ = a_0 - 1$ is a row maximizer. If the displayed value is zero, a_0 is the only other maximizer; otherwise a_* is the unique maximizer.*

Proof. All denominators in (1) are positive. With $t = n - 2a + 1$, direct expansion yields

$$2[a(a-1)(n-2a+3) - (n-a+1)^2(n-2a+1)] = H_n(t). \quad (2)$$

Thus $H_n(t)$ has the sign of $q_n(a) - 1$. Moreover,

$$H_n(t+2) - H_n(t) = -n(8t+14) - 2 < 0.$$

As a increases, t decreases by two, so the relevant values of $H_n(t)$ increase strictly. The first endpoint has the required negative sign because

$$H_n(n-3) = -2(n^3 - 5n^2 + 5n - 1) < 0.$$

At the last admissible index, $t = 2$ when n is odd and $t = 3$ when n is even; the corresponding values are

$$H_n(2) = n^2 - 14n - 3 > 0, \quad H_n(3) = n^2 - 27n - 4 > 0.$$

Hence a_0 exists. Before a_0 , equation (1) says that the row is strictly increasing. At and after a_0 , it is nonincreasing, with equality possible only at the first transition. This proves the claim. $\square$

3 The exact finite certificate

The following two identities allow a localized maximum to be transported between consecutive rows. They follow by cancellation of factorials:

$$\frac{R_{n+1}(a)}{R_n(a)} = \frac{(n+1)^2(n-1)(n-2a+2)}{2(2n-1)(n-a+1)^2(n-2a+1)}, \quad (3)$$

$$\frac{R_{n+1}(a+1)}{R_n(a)} = \frac{(n+1)^2(n-1)(n-2a)}{2(2n-1)a(a+1)(n-2a+1)}. \quad (4)$$

Proposition 3. *For every integer $496 \leq n \leq 100000$, one has $\rho(n) < 1$.*

Exact-computation proof. The accompanying reference program implements the following finite procedure. For each n , binary search on the strictly monotone sign sequence from Lemma 2 finds a_0 , and hence a genuine maximizing index $a_n = a_0 - 1$, using integer operations only. The initial value $R_{496}(a_{496})$ is computed directly from binomial coefficients as a reduced rational number. At each subsequent row, the program first determines a_{n+1} independently from H_{n+1} , verifies that $a_{n+1} - a_n \in \{0, 1\}$, and then applies (3) or (4). Each decision $R_n(a_n) < 1$ is the exact integer comparison of the numerator and denominator of a reduced fraction.

The program checked all $100000 - 496 + 1 = 99,505$ rows. Its audit summary is shown in Table 1. The transition counts sum to 99,504, as required. Direct recomputation from the defining binomial formula at $n = 496, 497, 2000, 2001, 4096, 10000, 50000, 100000$ agrees with recurrence propagation. The largest rational encountered occurs at $(n, a) = (497, 241)$; its serialized denominator-minus-numerator is a positive integer. Thus every localized maximum, and hence every value in each row, is strictly below one. The files `exact_scan.py`, `resolution_check.py`, and `resolution_check_results.json` provide the complete executable procedure and exact output.

Table 1: Exact finite-certificate summary.

Quantity	Exact result
Rows checked	99,505
Transitions with $a_{n+1} = a_n$	49,855
Transitions with $a_{n+1} = a_n + 1$	49,649
Rows with tied adjacent maxima	0
Rows with $\rho(n) \geq 1$	0
Direct-formula cross-checks	8 of 8 passed
Largest scanned value	$R_{497}(241) < 1$ exactly

□

No floating-point number enters this certificate. The reference implementation uses arbitrary-precision integers and exact rational reduction from the Python standard library; rerunning it either reproduces every asserted comparison or terminates at a failed assertion.

4 A uniform bound for the infinite tail

Fix $n \geq 100001$, choose the left maximizer $a_* = a_0 - 1$ from Lemma 2, and set

$$x = n - 2a_*.$$

The two signs adjacent to the first transition are

$$H_n(x+1) < 0 \leq H_n(x-1).$$

The second inequality is

$$0 \leq n^2 - n(2x^2 - x - 1) - x,$$

and therefore $2x^2 - x - 1 < n$. Since $x \geq 2$, this implies the strict localization bound

$$x^2 < n. \tag{5}$$

If the right maximizer a_0 is tied with a_* , it has the same value and requires no separate estimate.

We use the classical one-sided Stirling inequalities (in a slightly weaker form than Robbins' bounds [2])

$$\sqrt{2\pi r} (r/e)^r < r! < \sqrt{2\pi r} (r/e)^r e^{1/(12r)}, \quad r \geq 1. \quad (6)$$

If $m \geq 1$, $0 < k < m$, $p = k/m$, applying the upper bound to $m!$ and the lower bound to the two denominator factorials gives

$$\binom{m}{k} < \frac{\exp(mh(p) + 1/(12m))}{\sqrt{2\pi mp(1-p)}}, \quad (7)$$

where $h(p) = -p \log p - (1-p) \log(1-p)$. For $k = (m-x)/2$, put $u = x/m$. Then

$$\log 2 - h((1-u)/2) = \frac{1}{2}((1-u) \log(1-u) + (1+u) \log(1+u)) \geq \frac{u^2}{2};$$

indeed, the derivative of the middle expression is $\operatorname{artanh} u \geq u$. Hence

$$\binom{m}{(m-x)/2} < 2^m \sqrt{\frac{2}{\pi m(1-x^2/m^2)}} \exp\left(-\frac{x^2}{2m} + \frac{1}{12m}\right). \quad (8)$$

Using the opposite sides of (6) gives, for $M \geq 1$,

$$\binom{2M}{M} > \frac{4^M}{\sqrt{\pi M}} \exp\left(-\frac{1}{6M}\right). \quad (9)$$

Because $a_* = (n-x)/2$, the defining ratio is exactly

$$R_n(a_*) = \frac{2n(x+1)}{n+x} \frac{\binom{n}{(n-x)/2} \binom{n-2}{(n-2-x)/2}}{\binom{2n-2}{n-1}}. \quad (10)$$

Apply (8) with $m = n$ and $m = n-2$, and apply (9) with $M = n-1$. Define

$$A_n = \frac{n-1}{n(n-2)}, \quad E_n = \frac{1}{12n} + \frac{1}{12(n-2)} + \frac{1}{6(n-1)}.$$

After cancellation, and using $n/(n+x) < 1$, equation (10) gives

$$R_n(a_*) < \frac{4\sqrt{A_n}}{\sqrt{\pi}} \frac{(x+1)e^{-A_n x^2 + E_n}}{\sqrt{(1-x^2/n^2)(1-x^2/(n-2)^2)}}. \quad (11)$$

For $A > 0$ and $x \geq 0$,

$$(x+1)e^{-Ax^2} \leq \frac{1}{\sqrt{2eA}} + 1,$$

because $xe^{-Ax^2} \leq 1/\sqrt{2eA}$. Also, (5) allows us to set

$$C_n = \left((1-1/n) \left(1 - \frac{n}{(n-2)^2} \right) \right)^{-1/2}$$

and bound the remaining reciprocal square root in (11) by C_n . Consequently

$$R_n(a_*) < \frac{4}{\sqrt{\pi}} \left(\frac{1}{\sqrt{2e}} + \sqrt{A_n} \right) C_n e^{E_n}. \quad (12)$$

For real $n \geq 5$, A_n and E_n decrease. The two factors in the product defining C_n^{-2} increase, so C_n also decreases. Thus the right-hand side of (12) is at most its value at $N = 100001$. The following rational bounds are verified by positive integer cross-products:

$$\frac{4}{\sqrt{\pi}} < \frac{231}{100}, \quad \frac{1}{\sqrt{2e}} < \frac{429}{1000}, \quad \sqrt{A_N} < \frac{3163}{10^6}, \quad C_N < \frac{1000011}{10^6}, \quad e^{E_N} < \frac{1000004}{10^6}. \quad (13)$$

For completeness, the first estimate follows from $\pi > 3$; the second from $e > \sum_{j=0}^6 1/j! = 1957/720 > 2718/1000$. The next two are direct squared comparisons of positive rationals. For the last, exact cross-multiplication gives $E_N < 4/1000004$, and $e^u \leq (1-u)^{-1}$ for $0 \leq u < 1$. The accompanying checker records every one of these cross-products.

Substitution of (13) into (12) gives the strict, wholly rational estimate

$$\begin{aligned} R_n(a_*) &< \frac{231}{100} \frac{432163}{10^6} \frac{1000011}{10^6} \frac{1000004}{10^6} \\ &= \frac{24957787612296876183}{25000000000000000000} < 1. \end{aligned} \quad (14)$$

This proves $\rho(n) < 1$ for every $n \geq 100001$. Together with Proposition 3, it proves Theorem 1.

Reproducibility and assistance disclosure

The finite certificate and the rational endpoint checks are supplied as human-readable Python source and JSON output. They use only exact integer and rational arithmetic; the analytic inequalities are proved in the text rather than inferred from computation. Generative-AI assistance was used to organize the exposition and to check algebra against the exact evidence. A separate model-based review rederived the central identities and replayed the exact certificate; this is not a substitute for independent peer review. No language-model confidence score, floating-point experiment, or unverified generated calculation is used as mathematical evidence. The author takes responsibility for the statement and proof.

References

- [1] J. Gil et al. *Points of Maximal Traffic on a Grid with Obstruction*. 2026. arXiv: 2609.01562 [math.CO]. URL: <https://arxiv.org/abs/2609.01562>.
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